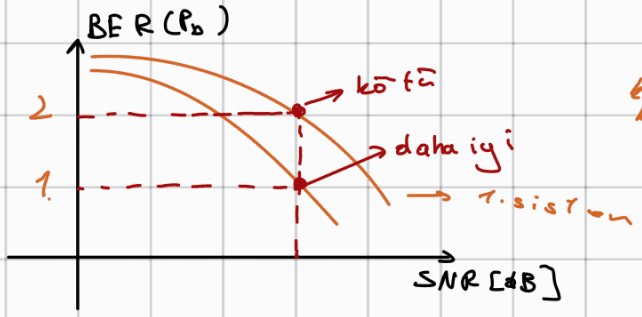
Shannon's Capacity Theorem

$C = W \log(1 + SNR)$

- C: kanal kapasitesi (bit/sn)
- W: kanalın genişliği (Hz)
- SNR: signal to noise ratio $\left(\frac{P_s}{P_n}\right)$

Bit error rate

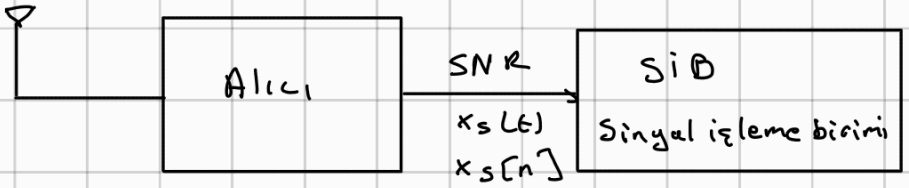


$C = W \log(1 + SNR)$

C: kanal kapasitesi (bit/sn), W: BG

RF ışıma yapar, anten çalıştığı bölgeye göre tasarlanmış olmalı.

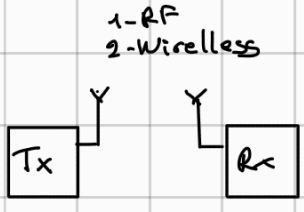
Antenna

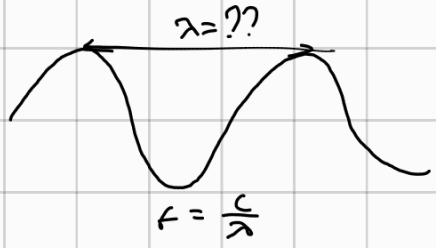


Gürültüyü minimalize etmeliyiz.

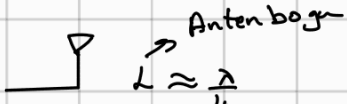
RF: 3khz frekansından 300 GHz'e kadar olan tüm singaller.

(100 GHz üstü miliniter wave, 10 GHz üstü microwave)





	f	$\lambda = \frac{c}{f}$	
VLF: (very low f.)	3-30khz	33km ~10km	Maritime radio navigation (Denizcilikte çok uzun mesafe haberleşme)
LF: (low f.)	30-300k	10km ~1km	
MF: (medium f.)	300k ~3M	1km ~100m	AM Radio aviation Navigation
HF:	3M ~30M	100m ~10m	?? radio
VHF:	30M ~300M	10m ~1m	→ TV, FM
UHF:	300M ~3G	1m ~100mm	→ TV, mobile, GPS, wifi, 4G
SHF:	3G ~ 30 G	100mm ~10mm	→ Wifi, satellite
EHF:	30G ~ 300G	10mm ~1mm	→ Astronomy satellite



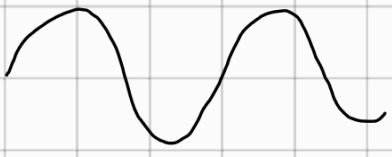
$L \approx \frac{\lambda}{4}$

$L = \frac{\lambda}{4}$

frekans büyüdükçe anten küçülür.

$f = \frac{c}{\lambda} \Rightarrow \lambda = \frac{c}{f}$

Noise

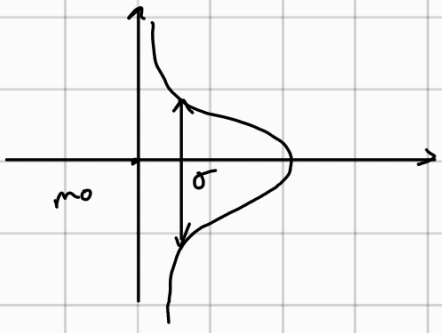


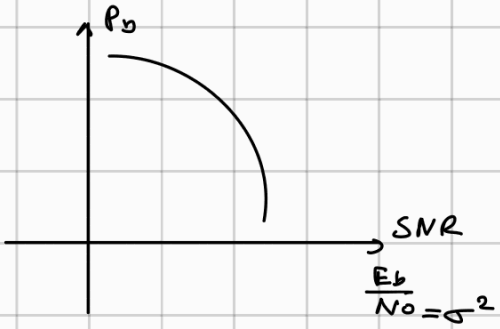
gürültü →



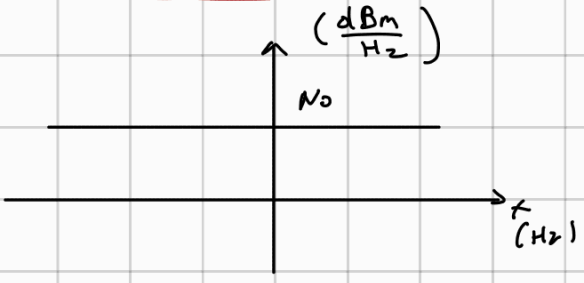
→ Rastgele süreç

varyans gücü

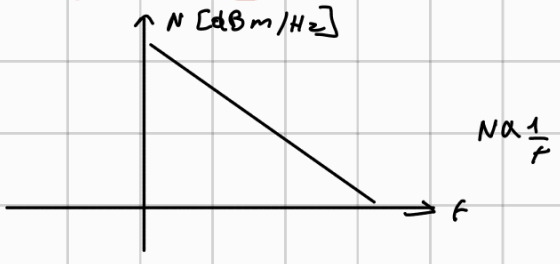




Whitenoise



Pink noise

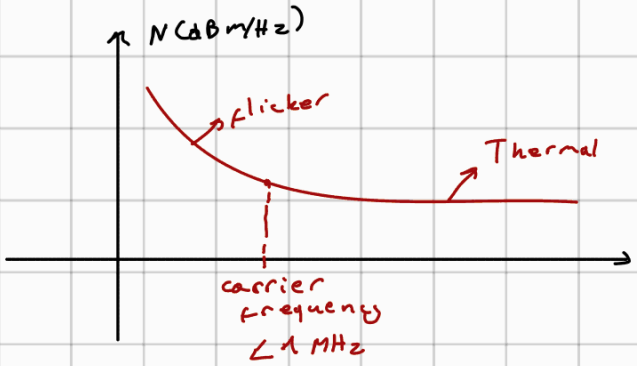


White Noise ~ Thermal Noise (Shot Noise) [→ White Noise N(m0,σ²)]

Flicker Noise ($\frac{1}{f^n}$ noise) ~ Pink noise

(Low frequency)

+ üst üste cierset



SNR oranını



$$SNR = \frac{P_s[W]}{P_n[W]} \qquad [\cdot]^{\text{birimsiz}}$$

log [SNR] birimi bel
10 log [SNR] decibel

$$SNR[dB] = 10\log\left[\frac{P_s}{P_n}\right] dB$$

SNR

$$10\log[2\cdot SNR] = \underbrace{10\log 2}_{3db} + 10\log[SNR]$$

2 kat arttırırsak 3db

10 kat arttırırsak 10db

100 kat " 20 db

SNR [dB] = 10 log [Ps/Pn]
10 log [2 SNR] = 10 log 2 + 10 log [SNR]
2 kat artarsa 3db artar

$$10\log\left[\frac{P_s}{1W}\right] \rightarrow dBW$$

$$10\log\left[\frac{P_s}{1mW}\right] \Rightarrow dBm$$

$10\log\left[\frac{P_s}{1W}\right] \Rightarrow dBW$

$10\log\left[\frac{P_s}{1mW}\right] \Rightarrow dBm$

10 log (10) = 10 dB
10 log (20) = 13 dB
10 log (40) = 16 dB
10 log (80) = 19 dB
10 log (100) = 20 dB

$$10\log[P_s] + 10\log\left[\frac{1}{\underbrace{1mW}_{10^{-3}W}}\right] = (P_s[dBw] + 30\,dB)\ (dBm)$$

$$\underbrace{10\log P_s}_{dBw} + \underbrace{10\log\left(\frac{1}{10^{-3}}\right)}_{30\,dB}$$

dBw + 30dB = dBm

dBw + 30dB = dBm

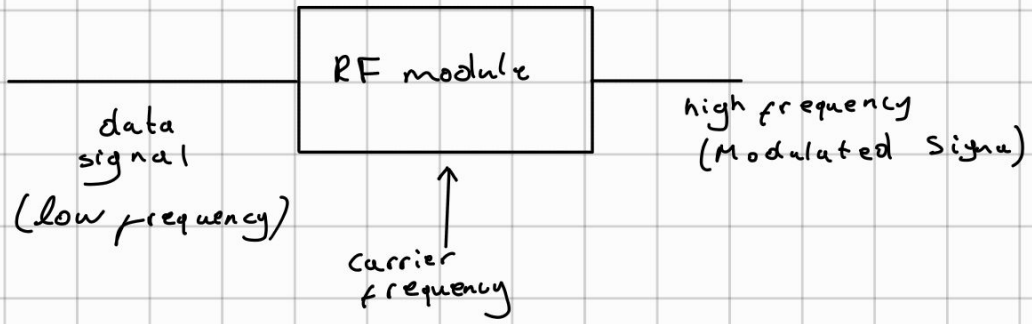
$$10\log\left[\frac{1mW}{1mW}\right] = 0\,dBm$$

$$10\log\left[\frac{10mW}{1mW}\right] = 10\,dBm$$

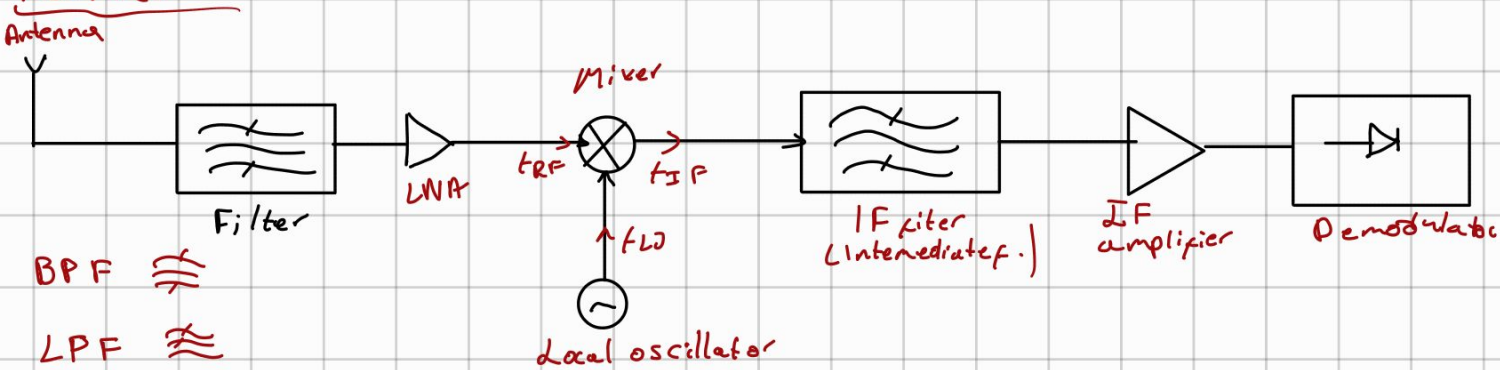
RF module

Receiver — Transmitter

Transreceiver



Reciver



BPF

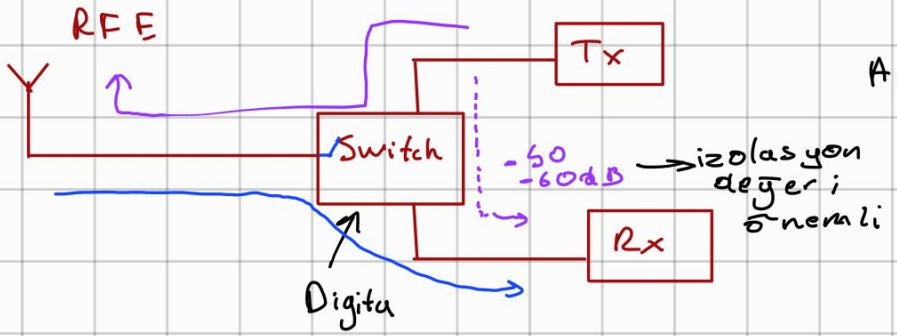
LPF

HPF

BSF

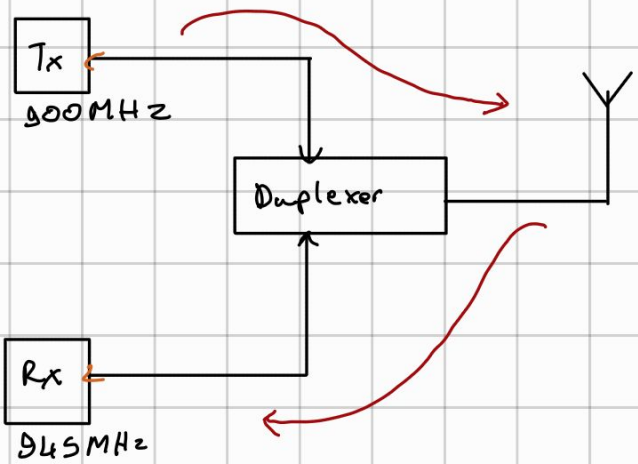
RF Front-End

RFE

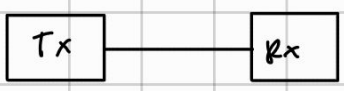


Aynı anda hem verici hemalıcı çalışmaz

Half duplex



Full duplex

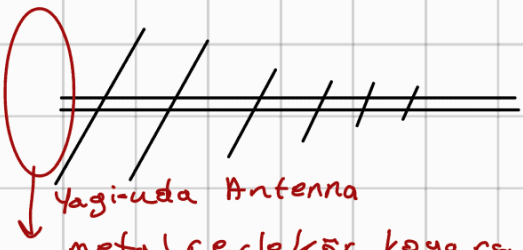


Simplex

Antennas

Log-periodic Antenna

heriki yerelon alır



Yagi-uda Antenna

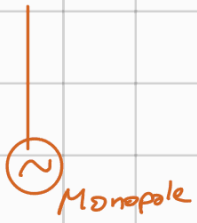
metal reflektör koyarsanız sadece tek yönlü ← olur



Bow Tie



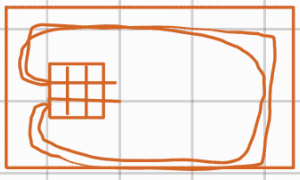
Dipole



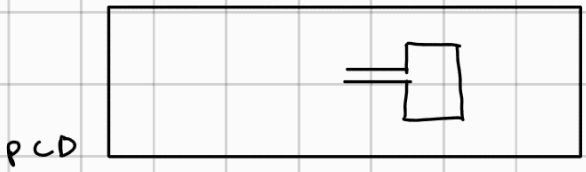
Monopole



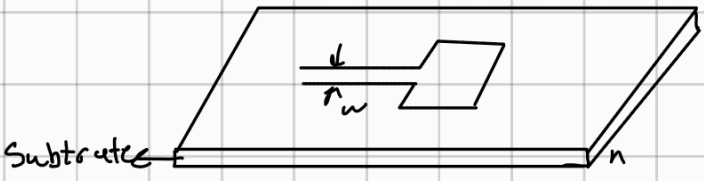
Circular



Patch Antenna



PCD

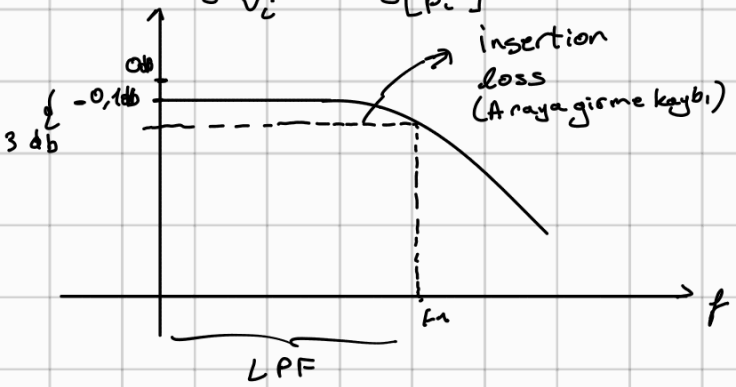


Substrate

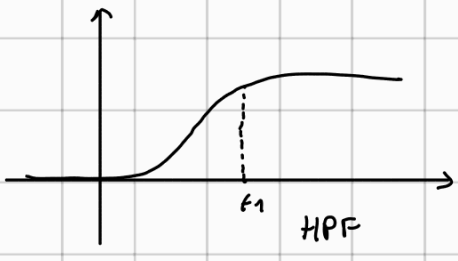
n

RF Filters

$20\log\frac{V_o}{V_i} = 10\log\left[\frac{P_o}{P_i}\right] = G$



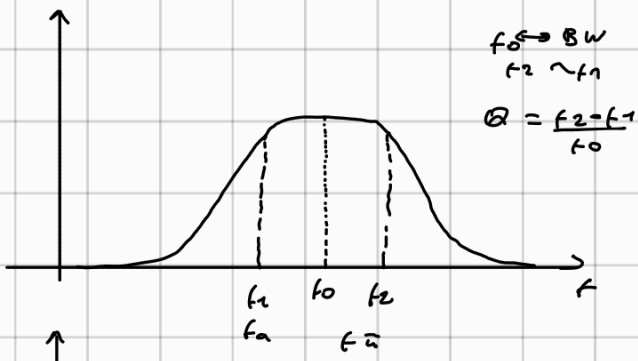
LPF

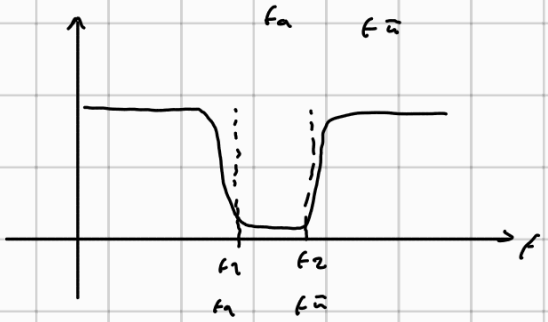


HPF



$BW = f_2 - f_1$

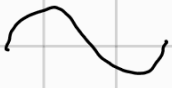


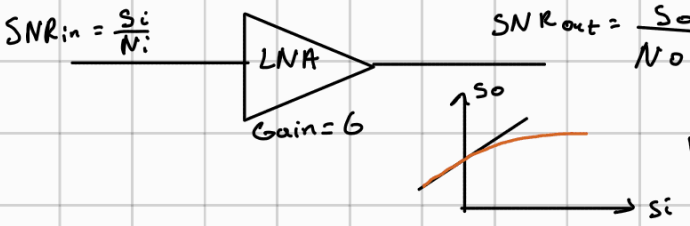


BSF

Aktif komponentler akımı yön veriyor.

LNA





$SNR_{out} = \frac{S_o}{N_o + N_A} = \frac{S_i \cdot G}{N_i G + N_A}$

↓ Filtrenin kendisinden kaynaklanan ek gürültü

N_A << seçilmeli

$SNR_{out} < SNR_{in}$

Requirements

- Low noise coming from device (⇒ low NF)
  - ↳ Noise Figure
- Linearity (⇒ high P_1dB → 1dB Compression Point
  - IP3 → Third order intercept Point)
- Low Power Consumption
- Low Area " (Small form factor)
- Gain (⇒ S_{21})
- Matching to anten filter